

Technical Specification: Microcircuit Assembly

Biological Blueprint Parsing and Normalization

Research and Development Documentation

March 26, 2026

1 Introduction

This document describes the `calculate_bbp_relative_presences` function, which serves as the biological “blueprint” parser for the microcircuit. It extracts raw population data from the Blue Brain Project (BBP) somatosensory cortex (S1) model and converts absolute cell counts into normalized relative percentages to ensure biological scaling across different microcolumn volumes.

2 Core Functionality: `calculate_bbp_relative_presences`

The primary goal of this function is to determine the compositional makeup of each cortical layer. Rather than relying on static counts, it calculates the proportional contribution of specific morphological types (m-types) to their respective biological groups (Excitatory or Inhibitory) within each layer.

2.1 1. Data Parsing & Categorization

The function processes the `S1-cells-distributions-Rat.txt` dataset, which typically contains five columns: *metype*, *mtype*, *etype*, *n*, and *m*.

- **Targeting M-Types:** The parser specifically extracts the 5th column (*m*), representing the total count for a broader morphological type across all its electrical variants.
- **Biological Mapping:** Utilizing an internal `acronym_map`, the function categorizes each m-type into one of two primary biological classes:
 - **Excitatory:** (e.g., Pyramidal Cells like `L23_PC`)
 - **Inhibitory:** (e.g., Martinotti Cells, Basket Cells like `L4_LBC`)

2.2 2. Layer Isolation

The function identifies cortical layers (*L1* through *L6*) by parsing the prefix of the m-type string (e.g., identifying “*L23*” from `L23_PC`). This allows the model to treat each layer as a distinct biological compartment, each with its own unique population totals and internal ratios.

2.3 3. Relative Percentage Calculation

Once raw counts are grouped by layer and biological type, the function performs a normalization calculation:

$$\text{Percentage} = \left(\frac{\text{Count of specific m-type}}{\sum \text{Counts of all peers in same Layer and Type}} \right) \times 100 \quad (1)$$

This normalization ensures that if the microcolumn radius is scaled up or down, the biological proportions remain identical to the anatomical data found in the rat S1 model.

3 Output Structure

The function returns two primary objects:

results A nested dictionary structured as [Layer] [BioType] [MType]. Each entry contains the absolute count and the calculated relative percentage.

layer_totals A summary dictionary containing the total number of Excitatory and Inhibitory neurons found within each layer.

4 Significance in Pipeline

This function is a critical prerequisite for `generate_microcolumn_cells`. By establishing these percentages first, the simulation can dynamically populate a 3D cylinder of any arbitrary radius while ensuring that the ratio of specific cell types (e.g., *L5_TTPC1* relative to other Layer 5 Excitatory cells) remains biologically accurate.

5 Usage Example

```
from blueprint_parser import calculate_bbp_relative_presences

# Path to the BBP distribution file
dist_file = "data/S1-cells-distributions-Rat.txt"

# Parse and normalize proportions
results, totals = calculate_bbp_relative_presences(dist_file)

# Example: Accessing L5 Excitatory Martinotti Cell percentage
l5_mc_pct = results['L5']['Inhibitory']['L5_MC']['percentage']
```